ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech MBA /B.Tech M.Tech/B.Tech (CSE)

Semester: VI

SUBJECT CODE: BCS601

COMPUTER NETWORKS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is the primary goal of computer networks?
(ii) What is Layered Architecture?
(iii) Name any two-transmission media used in networking.
(iv) What is the backbone of a network?
(v) What is the difference between unicast, multicast, and broadcast?

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is congestion control in networking?
(ii) How does transport layer perform duplication control?
(iii) List any two advantages of using IPv6 over IPv4.
(iv) What is the role of the presentation layer?
(v) Define the term "sliding window protocol.

SECTION – B

NOTE: Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) What do you understand by switching? Explain its types also in brief.

OR

(b) Discuss the role of the transport layer in network communication.

(ii)(a) Explain in detail about the error and flow control mechanisms employed at data link layer.

OR

(b) Discuss the OSI model in detail. Explain the functions of each layer with suitable examples.

Q.4.(i)(a) Compare and contrast guided and unguided media for transmission.

OR

(b) If a network has a bandwidth of 10Mbps and a latency of 50ms, what is the bandwidth-delay product?

(ii)(a) Find the data word for 1001110 using CRC decoder where generator polynomial is x3+x+1. Check whether word can be accepted or not.

OR

(b) Compare and contrast IPv4 and IPv6 addressing schemes. Discuss the reasons behind the transition from IPv4 to IPv6.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5. (a) List different carrier sense protocols. How CSMA/CD protocol is different from other CSMA/CA protocol?

OR

(b) A system uses a sliding window protocol with a window size of 4 and a transmission time of 2ms per packet. Calculate the total transmission time for 10 packets.

Q.6. (a) If transmission delay and propagation delay in a sliding window protocol are 1 msec. and 99.5 msec. respectively, then:

i. What should be the sender window size to get the maximum efficiency?
ii. What is the minimum number of bits required in the sequence number field?
iii. If only 7 bits are reserved for sequence numbers, then what will be the efficiency?

OR

(b) With a neat sketch explain the steps of Data Encryption Standard to encrypt plain text to cipher text.